LUSEE-NIGHT · FARADAY · COLLABORATION BRIEFING

The pixel sum never *converged*

Why the diffuse-Faraday estimator was replaced rather than repaired — the mechanism, the null tests that have power, and the answers to the three questions this always draws.

> The old estimator summed $w_n\ e^{2i\varphi_n\lambda^2}$ over HEALPix pixels. At LuSEE wavelengths the Faraday phase turns tens to hundreds of times *between adjacent pixels*, so that sum is not a sky integral — it is a two-dimensional random walk whose length is set by the pixel count.
>
> Its value falls as the grid is refined and never settles. What we published was the quadrature error, not the sky.

01 What changes, and what does not

The scope is narrow. Everything that does not involve a spatially structured Faraday screen is untouched — the finding is about one quadrature, not about the pipeline.

COMPONENT	STATUS	WHY
Stokes I → 16 products	stands	Band-limited, deterministic
I → Q,U leakage	stands	No Faraday phase involved
Point source at one φ_{FD}	stands	Single depth ⇒ no gradient
Polarimeter calibration	stands	Instrument-side
Spectrometer & zoom bins	stands	Instrument-side
Diffuse sky-pol through the screen	withdrawn	Phase unresolved by ~10^2–10^3

Diffuse sky pol through the screen	withdrawn	Phase unresolved by 10^{-4} – 10^{-3}
Step-4 diffuse delay spectrum	withdrawn	Equals the RM histogram

Concretely: the residual quoted as the sky-polarization contribution, $|dP|/I \approx 1.7 \times 10^{-4}$, and the amplitude of the Step-4 delay spectrum do not survive. The qualitative conclusion already in `PROGRESS.md` — that the observed P through the screen is leakage-dominated — survives and gets stronger.

02 An integral, and the sum that was standing in for it

The physical observable is a continuum integral over the sky,

what we want

$$P(\lambda^2) = \int w(\hat{n}) \cdot e^{2i\phi(\hat{n})\lambda^2} \, d\Omega$$

and the code evaluated it as an equal-area HEALPix quadrature — a midpoint rule by another name,

what we computed

$$\hat{P} = \Sigma_p \, w_p \cdot e^{2i\phi_p\lambda^2} \, \Delta\Omega$$

A quadrature rule converges to its integral *only if the sample spacing resolves the integrand*. That condition is not a technicality here; it is the entire story. The integrand is a unit-modulus phasor whose argument is $2\phi(\hat{n})\lambda^2$, and the Hutschenreuter map is smooth in ϕ but λ^2 at LuSEE is enormous.

Measured on the map itself: neighbouring nside-512 pixels differ in rotation measure by a median `0.56 rad m⁻²` (p90 `4.3`). Multiply by $2\lambda^2$ and that is the phase step between adjacent samples.

6.4
full turns of Faraday phase between median-neighbouring pixels at 50 MHz

18
the same at 30 MHz

160
the same at 10 MHz

< ½
turns per sample that Nyquist

actually allows

The estimator itself is not wrong — it is standard RM synthesis, and it is the right tool at the frequencies where the Faraday map was *measured*. Coherence across a pixel needs $2\,\Delta RM\,\lambda^2 \lesssim 1$, which on this map holds above about **317 MHz**. LuSEE-Night observes roughly a decade below that floor. The same estimator that builds the input map cannot be run on the data the map is being applied to.

03 The picture that settles it

An RM map is normally displayed as itself — smooth, structured, obviously physical. The quantity the simulation actually multiplies Q+iU by is $e^{2i\cdot RM\cdot\lambda^2}$, and that is a different object. Wrapping its phase into $[-\pi, \pi)$ is the honest way to look at what the grid is being asked to sample.

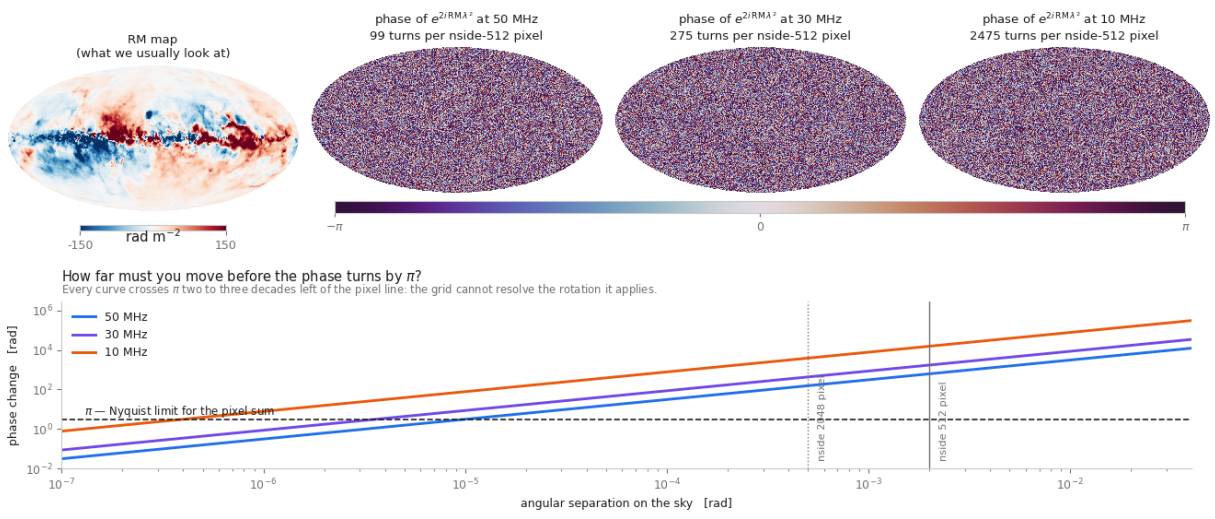


FIGURE A · AUDIT/FARADAY_PHASE_MAPS.PNG

Nothing survives of the structure on the left. The three phase maps are the same sky at 50, 30 and 10 MHz. The lower panel is the same statement quantitatively: every band crosses π two to three decades to the left of the pixel line — and moving from nside 512 to 2048 (the dotted line) buys two of the ten decades needed.

> HAVE THIS READY
>
> The turn counts in the figure titles (`99 / 275 / 2474`) come from the *rms* map gradient, `4328 rad m⁻² per radian` → `8.65 rad m⁻²` per nside-512 pixel. The report quotes `6.4 / 17.8 / 160`, from the *median* adjacent-pixel difference of `0.56 rad m⁻²`. Both are

> correct; the gradient distribution is heavy-tailed, so its rms sits ~15× above its median (the median $|\nabla RM|$ gives 0.83 per pixel, which reconciles the two). **Quote the median** — it is the conservative number, and it is still 18 turns per pixel at 30 MHz.

04 Why random phases mean a random walk

This is the question that always comes. The short answer is: **yes — the HEALPix sum does not converge to the integral**, and the reason random-looking samples do not average into an integral is worth being precise about.

Nothing is actually random

Run the simulation twice and you get the same number to the last bit. "Random" here means *equidistributed and uncorrelated between neighbours*, which is what happens to any smooth function sampled far below its own oscillation scale. It is the same mechanism as aliasing: sample $\sin(2\pi f t)$ at $\Delta t \gg 1/f$ and the sequence you get has no memory from one sample to the next, even though the function is perfectly deterministic. Eighteen turns per pixel puts $\theta_p = 2\varphi_p\lambda^2 \mod 2\pi$ in exactly that regime.

So the sum is a walk, not an average

Once the directions are uncorrelated, $\Sigma\ w_p\ e^{i\theta_p}$ is literally a walk in the complex plane: N steps, step p of length $|w_p|$ in an arbitrary direction. Every cross term averages to zero over the circle, so

$$\langle |P|^2 \rangle \;=\; \Sigma_{n,m}\ |w_n||w_m|\ \langle e^{i(\theta_n-\theta_m)} \rangle \;=\; \Sigma_n\ |w_n|^2$$

the expected squared length of the walk. The beam normalisation divides by $\Sigma|w_n|$, so the reported amplitude is $\sqrt{(\Sigma|w|^2)}/\Sigma|w|$ — which for equal weights is $N^{-1/2}$, and in general is $N_{\mathrm{eff}}{}^{-1/2}$. **That is a number about the grid. There is no sky in it.**

Why refining does not rescue it

Think of the discrete sum as Monte-Carlo integration, because that is what it has become. Sampling a function at N points estimates its mean with an error $\sim\ \sigma_f/\sqrt{N}$, where σ_f is the function's spread over the domain. Here the integrand is a phasor pointing anywhere, so $\sigma_f = O(1)$, while the true mean is tiny — a fast-oscillating integrand against a smooth weight very nearly cancels. The estimator is therefore

$$\hat{P} = P_{true} + \textbf{quadrature error}, \quad \text{with} \quad |\text{error}| \gg |P_{true}|$$

and it is dominated by its own error term. Refining the grid shrinks that error as $N_{eff}^{-1/2}$ — so the answer keeps falling and never settles — until the pixel scale finally resolves the phase, at which point the error drops below P_{true} and the sum plateaus at the physical value. Everywhere in the LuSEE band we are on the falling branch: the incoherent floor is 80–150% of the measured value at 30 MHz and 120–190% at 10 MHz, i.e. the value *is* the floor.

The name *pixelization shot noise* is exact by analogy with the $1/\bar{n}$ discreteness term in a galaxy power spectrum: a positive, additive floor set purely by sampling density, carrying no information about the field's correlations.

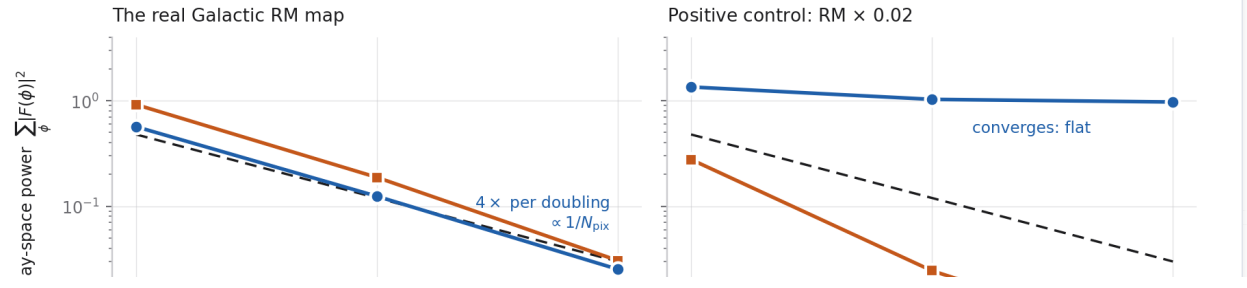
> THE ONE-LINE VERSION
>
> A Riemann sum only converges when it resolves its integrand. This one is **two to three decades past Nyquist**, so it converges to nothing — it just reports its own Monte-Carlo error, which happens to be a smooth, plausible-looking function of frequency.

05 The null test that has power

"The result looks like noise" is an assertion until the test is shown to be capable of saying otherwise. So: permute the RM values across pixel indices with a fixed seed. The histogram is preserved exactly, every spatial correlation is destroyed, and the weights are untouched. If the result carried sky structure, this would wreck it.

It does not. And the same test, run on a sky scaled down until the pixel sum *does* converge, separates without bound — so the null result means something.

Destroying every spatial correlation in the RM map changes nothing — unless the sky is weak enough to converge



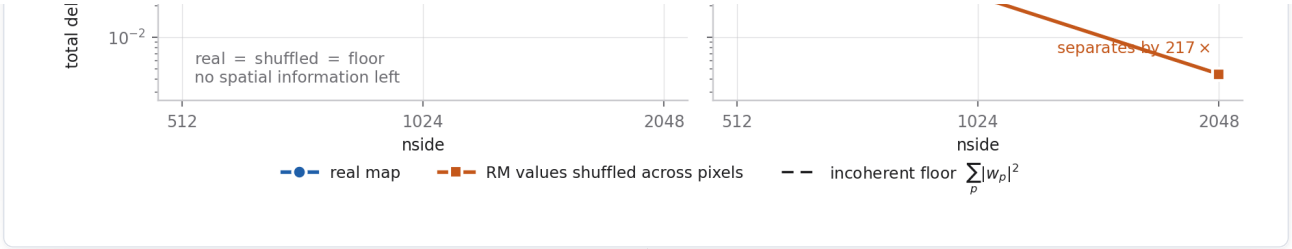


FIGURE B · FROM AUDIT TEST 06

Left: on the real sky, real map, shuffled map and the parameter-free incoherent floor $\Sigma|w_p|^2$ are the same curve, and it falls a factor 4 per nside doubling — $1/N_{pix}$. The real/shuffled ratio is `0.62, 0.66, 0.83` at nside 512, 1024, 2048. **Right:** the identical test on `RM × 0.02`, which pushes the sky into the converged regime. Now the real map plateaus and its shuffle collapses — a separation of `4.9×, 42×, 217×`. The test has power; the null result is informative.

Two companion checks point the same way. The delay spectrum agrees with the $|w|^2$-weighted *histogram* of the RM map to 4% — one line of `np.histogram`, no beam convolution and no phases — so every feature we had reported was a property of the map alone: the "envelope imaging the sky RM distribution" *is* the histogram; the "cutoff at $\varphi \approx 2400$" is the largest $|RM|$ above the horizon; the ν^{-3} scaling is the $d\lambda^2/d\nu$ Jacobian. And smoothing only the Q/U maps to 1°, 3° and 10° — which *would* show up, because the delay transform refocuses pixels of equal RM — leaves the measured/incoherent ratio flat at `1.0` at every scale. There is no coherent excess to find.

06 The slope is not sloppy $1/\sqrt{N}$ – it is exactly $1/\sqrt{N_{eff}}$

Figure 2 of the report is the summary, and the honest reaction to it is that the decay does not cleanly track the dotted $N_{pix}^{-1/2}$ guide. Measured log-slopes in N_{pix} are `−0.33` to `−0.38` against `−0.50`, and the 64→512 drop is 3.6× rather than 8×. Two separate things are going on, and neither is a defect.

The dotted line is the wrong prediction

$N^{-1/2}$ assumes equal-weight steps. The actual weight is $|w|^2$ = beam × polarized sky, which is *extremely* concentrated: 5.1% of all weight sits in the top 0.01% of pixels, and Taurus A alone carries 4.0% of it in 0.008% of the sky. The right prediction is $\sqrt{(\Sigma|w|^2)}/\Sigma|w| = N_{eff}^{-1/2}$ with $N_{eff} = (\Sigma w)^2/\Sigma w^2$ — and because the weight map is itself unresolved at nside 64–256, N_{eff} grows only as $N_{pix}^{0.80}$. That gives a predicted slope of `−0.40`.

And a single λ² sample is Rayleigh-distributed

~~And a single λ² sample is Rayleigh-distributed~~

Report Fig. 2 evaluates the sum at *one* wavelength. For a 2-D random walk `|P|²` is exponentially distributed, so a single sample carries 52% fractional scatter about the rms. The bounce — including the non-monotonic nside 256→512 leg — is exactly that.

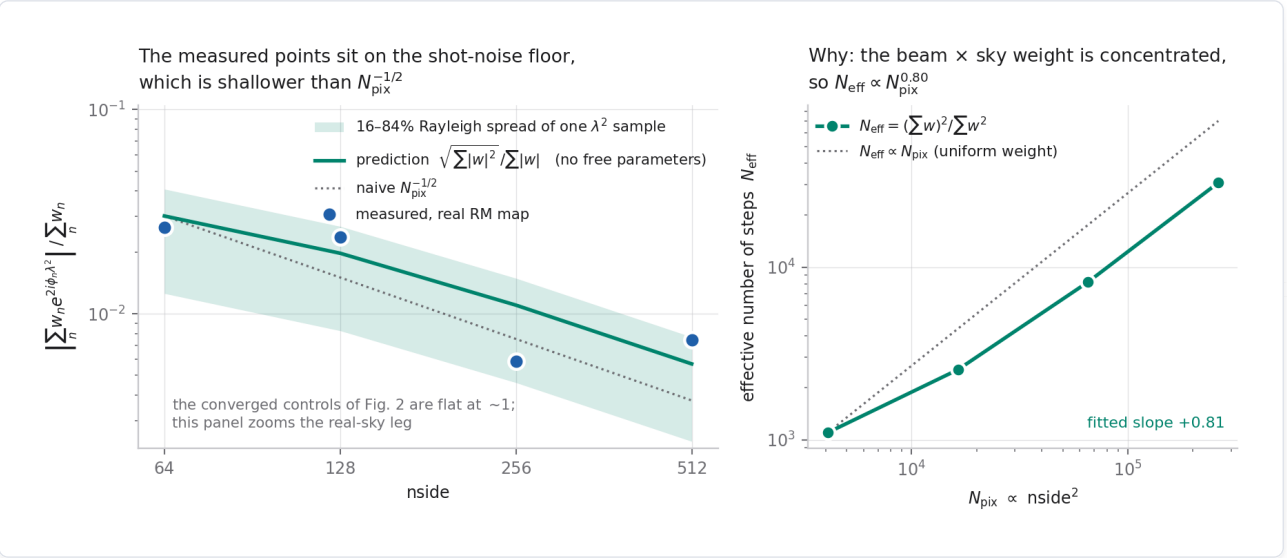


FIGURE C · NEW, FROM STEP5_INTUITION.NPZ

Left: the same measurement as Fig. 2's real-RM leg, against the zero-parameter floor `√(Σ|w|²)/Σ|w|` and the 16–84% Rayleigh spread of a single λ² sample. Measured/predicted is `0.88, 1.21, 0.53, 1.31` — all four inside the band. The naive $N_{pix}^{-1/2}$ guide is steeper than the data because it is the wrong prediction, not because the data are misbehaving. **Right:** why — N_{eff} runs 1100 → 30 900 over nside 64 → 512, a slope of `+0.81` where uniform weights would give `+1.00`.

> WHY THE AUDIT'S NUMBERS LOOKED CLEANER
>
> The audit ran a smooth `cos²θ` lobe, which *is* resolved at every nside, so there $N_{eff} \propto N_{pix}$ exactly — and band-averaging over 16 384 channels removed the Rayleigh scatter. That test gave `3.41, 4.57, 4.92` per doubling against a predicted 4.00. Same physics; the two figures differ only in weight map and in whether one λ² sample or the whole band is used.

So: is this a case for more nsides? Two different questions hide in that. For a *cleaner demonstration* — no: the fix is band-averaging, not refinement, and the audit already ran nside 256→2048 that way. For a *correct calculation* — also no, and that is the next section.

07 Three reasons brute force cannot fix it

Any one of these is sufficient on its own, and all three land in the same place — between roughly 350 and 800 MHz — by unrelated mechanisms.

1 · Resolution

Convergence needs $2|\nabla\varphi|\ \lambda^2\ \theta_{pix} < \pi$. Solving for the LuSEE band gives **nside ≈ 2.8 × 10^5 at 30 MHz** and **2.5 × 10^6 at 10 MHz** — of order 10^{12} and 10^{14} pixels. This is not a cluster-hours problem. And the payoff is negative: extrapolating the converged branch downward suggests the true suppression is *below* what the sim reports, not above.

2 · The screen is not determined

The observable depends on the map only through $e^{2i\varphi\lambda^2}$, so the map's own error becomes phase error. `faraday2020v2` ships per-pixel σ with a median $\sigma/|RM| = 0.50$, and on 24.6% of the sky the reconstruction does not fix even the *sign*. Propagated to phase, the median uncertainty is `312 turns` at 30 MHz and 2810 at 10 MHz, with **100% of the sky exceeding one full turn**. A perfect simulation of this map would be a perfect simulation of the wrong sky — and there is no better map; the uncertainty is a property of the reconstruction.

3 · The polarized template is not determined either

The floor $\Sigma|w_p|^2$ sums per-pixel magnitudes, so it weights every scale equally and noise-dominated small scales enter at full strength. WMAP K per-pixel polarization S/N is 1.45; roughly a third of the floor is WMAP's own instrument noise being Faraday-rotated. The sky model then carries that template from 23 GHz to 30 MHz at $\beta = -2.8$, a factor 1.2×10^8.

Taken together: inside the LuSEE band the diffuse deterministic calculation is not expensive, it is **ill-posed**.

08 What replaces it — and why the two regimes are disjoint

The natural objection is "then just work where it converges." Figure 2 answers that directly, by scaling the RM map down until the coherent sum *does* converge — and showing that where it converges, there is nothing to measure.

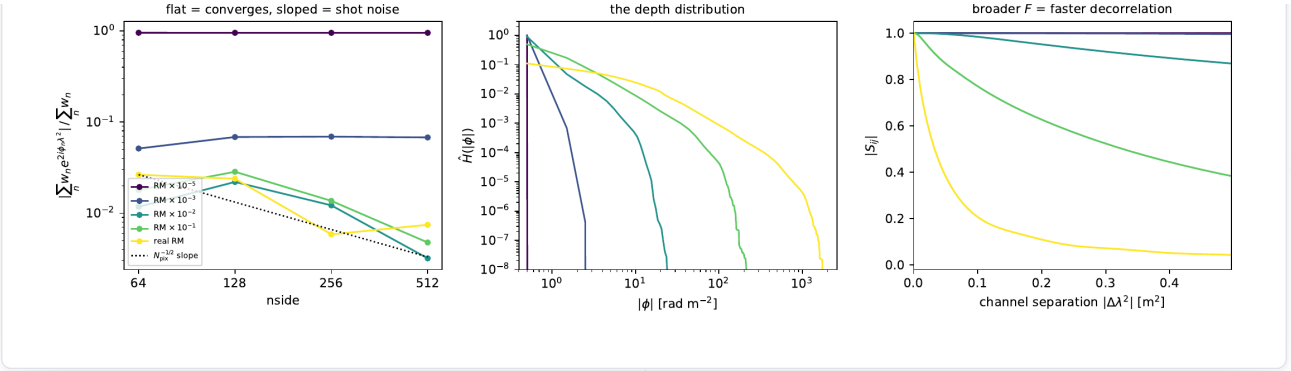


FIGURE 2 OF THE REPORT · STEP5_BRIDGE.PDF

Reading one colour left to right is one sky. The coherent sum converges only for RM × 10^{-5} and × 10^{-3} — and for those skies the whole depth distribution (middle) has collapsed to $\varphi \approx 0$, inside the region that must be discarded as degenerate with smooth instrumental leakage, and the channel covariance (right) is flat: no signal. Intrinsic angles are set to zero, so any decay in the left panel is the Faraday phase decorrelating, not the sky's own angles.

The replacement inverts the order of operations. **Bin in depth first, transform second.** Each pixel deposits its weight into the bin holding its Faraday depth, building a depth distribution $F(\varphi)$, and the transform is taken of *that*. Because F is a one-point statistic — it sees the joint distribution of (weight, depth) over pixels and nothing about their arrangement — it has no oscillatory integrand to alias, and it converges. The acceptance gates make that quantitative: the normalised $F(\varphi)$ moves by Kolmogorov distances of 2×10^{-5} to 1.9×10^{-4} across nside 256–2048 and under a rigid null rotation, against a 10^{-2} tolerance — while the coherent observable collapses by 134× under the same refinement and moves by 7.2× under the rotation alone.

> BE UPFRONT ABOUT THE COST
>
> Discarding the arrangement is exactly what makes F converge, and exactly what it gives up: the RM map's spatial power spectrum is thrown away, so the coherence angle θ_c cannot be recovered from the template. That is why the deliverable is reframed from *shape* to *detection*, and why the amplitude is quoted as a bracket with its reasons — $A_{upper} \approx 9.9\times10^{-3}$, $A_{slab} \approx 4.3\times10^{-4}$, $A_{disp} \approx 5.2\times10^{-7}$ at 30 MHz — never as a single value. Only A_{upper} contains θ_c.

09 If someone asks

Is the sum coded wrong?

No. It is computed correctly and it is the standard RM-synthesis estimator. It is being used two to three decades outside its domain of validity — above about 317 MHz on

this map it is the right tool, and it is what built the input map in the first place.

Doesn't a random walk still average to the right answer with enough samples?

It averages to the right answer at rate $1/\sqrt{N}$, and the true answer is far below that error for any N we can reach. You would need ~10^{12} pixels at 30 MHz before the estimate is signal rather than noise. Until then, refining does not converge the answer — it shrinks the answer, because the error *is* the answer.

Could this be Monte-Carlo noise, or map noise, rather than pixelization?

Nothing in the run is stochastic — it reproduces bit-for-bit. WMAP map noise does contribute, about a third of the floor, but smoothing the polarization to 10° removes it and the floor persists unchanged. The signature that identifies it is the scaling: a factor 4 per nside doubling, matched by a parameter-free prediction.

The trend in Fig. 2 isn't a clean $1/\sqrt{N}$ — doesn't that undercut the claim?

It strengthens it. The clean $N^{-1/2}$ line is the equal-weight idealisation; the real beam × sky weight is concentrated enough that $N_{eff} \propto N_{pix}^{0.80}$, predicting -0.40. All four measured points sit inside the 16–84% Rayleigh band of that zero-parameter prediction (Figure C). And the converged controls in the same panel are dead flat — slope -0.001 — so the two regimes are not ambiguous.

Is the broadband Faraday ripple real physics, then?

Yes, and nothing here says otherwise. A ripple from many superposed RMs is real. The claim is only that *this calculation was not measuring it*, and that the audit deliberately does not locate the true amplitude — the runs show the falling branch everywhere in-band and never reach the plateau.

Where does this live, so people can check it?

The full numerical audit — seven tests, each with its own control — is `audit/README.md` with reproduction scripts in `audit/scripts/`. The refuted results and their write-up are preserved at the `audit-2026-08-18` tag; cite the tag, not a branch.

10 Numbers to have ready

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QUANTITY	10 MHZ	30 MHZ	50 MHZ
λ^2 [m²]	898.8	99.9	36.0
Phase turns per nside-512 pixel (median)	160	17.8	6.4
… same, from the rms gradient	2474	275	99
nside needed for convergence	2.5×10^6	2.8×10^5	–
Median RM-map phase uncertainty (turns)	2810	312	112
Incoherent floor / |P| at nside 1024	1.18	1.55	1.44

SINGLE-NUMBER FACTS	VALUE
Median adjacent-pixel ΔRM at nside 512	0.56 rad m⁻² (p90 4.3)
Coherent sum valid above	317 MHz
Coherent power collapse, nside 256 → 2048	134× (predicted 64×)
Coherent observable under a rigid null rotation	7.2× spread
Normalised F(φ) under the same tests	KS 2×10^{-5} – 1.9×10^{-4}
Real / shuffled RM map, nside 512 / 1024 / 2048	0.62 / 0.66 / 0.83
Same, for the RM × 0.02 control	4.9 / 42 / 217
Delay spectrum vs. the plain RM histogram	ratio 1.04
Q/U smoothing sweep, 0° → 10°	ratio flat at 1.0
Sky fraction where σ_{RM} > |RM|	24.6%
WMAP K per-pixel polarization S/N	1.45
Weight in the top 0.01% of pixels	5.1%

Sources: `audit/README.md` (2026-08-18 numerical audit, tests 01–07); `report/faraday_depth_template/faraday_depth_template.tex` §§2–3 and the acceptance gates;

`laladdy_depth_template`, `laladdy_depth_template.tex.jj2` and the acceptance gates; `generated_data/step5_intuition.npz` and `audit/generated/shuffle_results.json`. Figures A and 2 are existing products; Figures B and C were generated for this briefing from the stored results. Refuted material is preserved at the `audit-2026-08-18` tag.